

Fairness-Preserving Lightweight Deepfake Detection

Full Training & Evaluation Pipeline

This notebook automates every step of the research pipeline:

1. Environment & GPU check
2. Dataset statistics & demographic distribution
3. Teacher training (XceptionNet)
4. Baseline student training (MobileNetV2)
5. Fairness-aware distillation (our method)
6. Standard distillation (ablation)
7. Comprehensive evaluation + fairness metrics
8. Cross-dataset generalization
9. Ablation comparison tables
10. Per-group accuracy & ROC charts
11. Grad-CAM heatmaps
12. Model size & speed profiling

Cell 0 — Setup & Imports

```
In [1]: import os, sys, time, warnings
warnings.filterwarnings('ignore')

# Add project root to path
PROJECT_ROOT = r'D:\M3 Projects\DeepFake_Research'
sys.path.insert(0, PROJECT_ROOT)
os.chdir(PROJECT_ROOT)

import torch
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import seaborn as sns
from IPython.display import display, HTML
from torch.utils.data import DataLoader

from src.config import (
    DEVICE, SPLITS_DIR, MODELS_DIR, RESULTS_DIR, FIGURES_DIR, GRADCAM_DIR,
    DEMOGRAPHIC_GROUPS, BATCH_SIZE, NUM_WORKERS,
    DISTILL_EPOCHS, STUDENT_EPOCHS, TEACHER_EPOCHS,
    ALPHA, BETA, GAMMA, DISTILL_TEMPERATURE
)
from src.models.xception import build_teacher
```

```

from src.models.mobilenetv2 import build_student
from src.data.dataset import create_data loaders, DeepfakeDataset
from src.evaluation.evaluate import evaluate_model, save_results, compare_models
from src.evaluation.cross_dataset import run_full_cross_dataset_evaluation

# Plot style
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette('husl')
COLORS = sns.color_palette('husl', 8)

os.makedirs(FIGURES_DIR, exist_ok=True)
os.makedirs(RESULTS_DIR, exist_ok=True)
print('✅ Imports OK')

```

✅ Imports OK



Cell 1 — GPU Check & Config

```

In [2]: print('='*55)
print(' ENVIRONMENT')
print('='*55)
print(f' PyTorch:      {torch.__version__}')
print(f' CUDA:         {torch.version.cuda}')
print(f' Device:         {DEVICE}')

if torch.cuda.is_available():
    gpu = torch.cuda.get_device_properties(0)
    print(f' GPU:           {gpu.name}')
    print(f' VRAM:           {gpu.total_memory/1e9:.1f} GB')
    print(f' CUDA cores:    {gpu.multi_processor_count} SMs')
else:
    print(' ⚠️ No GPU – training will be slow!')

print('='*55)
print(' TRAINING CONFIG')
print('='*55)
print(f' Batch size:     {BATCH_SIZE}')
print(f' Distill α:       {ALPHA} (distillation loss weight)')
print(f' Distill β:       {BETA} (fairness loss weight)')
print(f' Distill γ:       {GAMMA} (classification loss weight)')
print(f' Temperature:    {DISTILL_TEMPERATURE}')
print(f' Groups:         {DEMOGRAPHIC_GROUPS}')

# Stability settings for RTX 4050 on Windows
torch.backends.cudnn.benchmark = False
torch.backends.cudnn.deterministic = True
print('cuDNN deterministic mode: ON')

```

```

=====
ENVIRONMENT
=====
PyTorch:      2.6.0+cu124
CUDA:         12.4
Device:       cuda
GPU:          NVIDIA GeForce RTX 4050 Laptop GPU
VRAM:         6.4 GB
CUDA cores:   20 SMs
=====

TRAINING CONFIG
=====
Batch size:    32
Distill  $\alpha$ : 0.7 (distillation loss weight)
Distill  $\beta$ :  0.2 (fairness loss weight)
Distill  $\gamma$ : 0.1 (classification loss weight)
Temperature:   4.0
Groups:        ['Male-White', 'Male-Black', 'Male-Asian', 'Male-Other', 'Female-White', 'Female-Black', 'Female-Asian', 'Female-Other']
cuDNN deterministic mode: ON

```



Cell 2 — Dataset Statistics & Demographic Distribution

```

In [3]: splits = {}
for split in ['train', 'val', 'test']:
    path = os.path.join(SPLITS_DIR, f'{split}.csv')
    splits[split] = pd.read_csv(path)

print('Split sizes:')
for name, df in splits.items():
    real = (df['label']=='REAL').sum()
    fake = (df['label']=='FAKE').sum()
    print(f' {name:6s}: {len(df):>8,} total ({real:,} real / {fake:,} fake)')

# Cross-dataset sizes
for ds in ['Celeb-DF', 'DFD']:
    p = os.path.join(SPLITS_DIR, f'{ds}_test.csv')
    if os.path.exists(p):
        df = pd.read_csv(p)
        print(f' {ds:8s}: {len(df):>8,} total (cross-dataset test)')

train_df = splits['train']

fig, axes = plt.subplots(1, 3, figsize=(18, 5))
fig.suptitle('Training Set - Demographic Distribution', fontsize=14, fontweight='bold')

# 1. Group distribution
grp_counts = train_df['demographic_group'].value_counts().sort_index()
axes[0].bar(grp_counts.index, grp_counts.values, color=COLORS)
axes[0].set_title('Samples per Demographic Group')
axes[0].set_xlabel('Group')
axes[0].set_ylabel('Count')
axes[0].tick_params(axis='x', rotation=45)

```

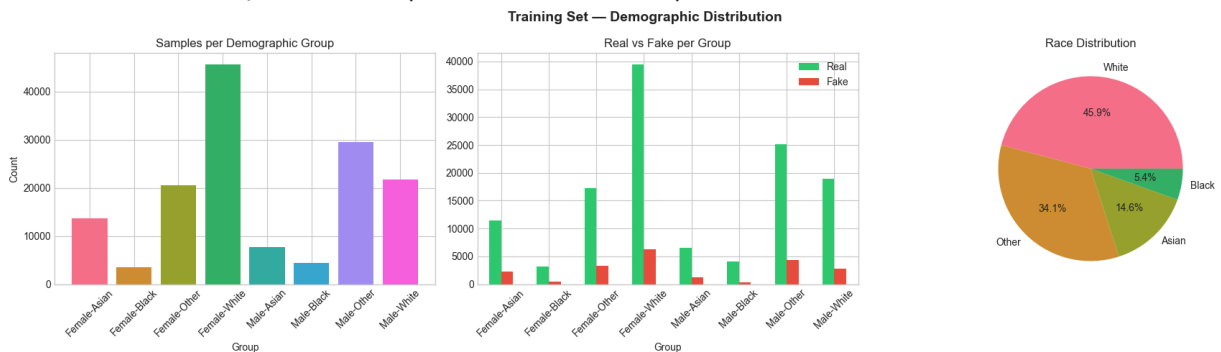
```
# 2. Real vs Fake per group
grp_label = train_df.groupby(['demographic_group', 'label']).size().unstack(fill_value=0)
grp_label.plot(kind='bar', ax=axes[1], color=['#2ecc71', '#e74c3c'])
axes[1].set_title('Real vs Fake per Group')
axes[1].set_xlabel('Group')
axes[1].tick_params(axis='x', rotation=45)
axes[1].legend(['Real', 'Fake'])

# 3. Race & Gender pie charts
race_counts = train_df['race'].value_counts()
axes[2].pie(race_counts.values, labels=race_counts.index, autopct='%1.1f%%',
            colors=COLORS[:len(race_counts)])
axes[2].set_title('Race Distribution')

plt.tight_layout()
plt.savefig(os.path.join(FIGURES_DIR, 'dataset_distribution.png'), dpi=150, bbox_inches='tight')
plt.show()
print('✅ Dataset stats plotted')
```

Split sizes:

```
train : 146,871 total (20,970 real / 125,901 fake)
val   : 31,487 total (4,500 real / 26,987 fake)
test  : 31,474 total (4,527 real / 26,947 fake)
Celeb-DF: 195,829 total (cross-dataset test)
DFD    : 102,359 total (cross-dataset test)
```



✅ Dataset stats plotted



Cell 3 — Train Teacher (XceptionNet)

Skips automatically if checkpoint already exists.

```
In [4]: from src.training.train_baseline import train_baseline

TEACHER_WEIGHTS = os.path.join(MODELS_DIR, 'xception_teacher_best.pth')
TEACHER_EPOCHS_NB = 20 # ← change as needed
TEACHER_BATCH = 64 # ← optimized for RTX 4050 6GB

if os.path.exists(TEACHER_WEIGHTS):
    print(f'✅ Teacher checkpoint found – skipping training.')
    ckpt = torch.load(TEACHER_WEIGHTS, map_location=DEVICE, weights_only=False)
    print(f'    Saved at epoch {ckpt["epoch"]+1}, val AUC = {ckpt["val_auc"]:.4f}')
else:
    print(f'🚀 Training Teacher (XceptionNet)...')
```

```

teacher = build_teacher(pretrained=True, device=DEVICE)
loaders = create_dataloaders(
    os.path.join(SPLITS_DIR, 'train.csv'),
    os.path.join(SPLITS_DIR, 'val.csv'),
    batch_size=TEACHER_BATCH,
    num_workers=0
)
optimizer = torch.optim.AdamW(teacher.parameters(), lr=1e-4, weight_decay=1e-5)
scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=TEACHER_EPOCHS)
results = train_baseline(
    teacher, loaders['train'], loaders['val'],
    optimizer, scheduler,
    num_epochs=TEACHER_EPOCHS_NB,
    model_name='xception_teacher',
    device=DEVICE
)
print(f'\n✅ Teacher training done. Best AUC: {results["best_auc"]:.4f}')

```

✅ Teacher checkpoint found – skipping training.
 Saved at epoch 11, val AUC = 0.9671

Cell 4 — Train Baseline Student (MobileNetV2, no distillation)

Skips if checkpoint exists. This is our ablation baseline.

```

In [5]: BASELINE_WEIGHTS = os.path.join(MODELS_DIR, 'mobilenetv2_baseline_best.pth')
BASELINE_EPOCHS = 50 # ← change as needed
BASELINE_BATCH = 128 # ← optimized for RTX 4050 6GB

if os.path.exists(BASELINE_WEIGHTS):
    print(f'\n✅ Baseline student checkpoint found – skipping training.')
    ckpt = torch.load(BASELINE_WEIGHTS, map_location=DEVICE, weights_only=False)
    print(f'    Saved at epoch {ckpt["epoch"]+1}, val AUC = {ckpt["val_auc"]:.4f}')
else:
    print(f'\n🔥 Training Baseline Student (MobileNetV2, no distillation)...')
    baseline_student = build_student(pretrained=True, device=DEVICE)
    loaders = create_dataloaders(
        os.path.join(SPLITS_DIR, 'train.csv'),
        os.path.join(SPLITS_DIR, 'val.csv'),
        batch_size=BASELINE_BATCH,
        num_workers=0
    )
    optimizer = torch.optim.AdamW(baseline_student.parameters(), lr=1e-4, weight_decay=1e-5)
    scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=BASELINE_EPOCHS)
    results = train_baseline(
        baseline_student, loaders['train'], loaders['val'],
        optimizer, scheduler,
        num_epochs=BASELINE_EPOCHS,
        model_name='mobilenetv2_baseline',
        device=DEVICE
    )
    print(f'\n✅ Baseline training done. Best AUC: {results["best_auc"]:.4f}')

```

🚀 Training Baseline Student (MobileNetV2, no distillation)...

[Student] MobileNetV2 loaded:

Total params: 2.6M

Trainable params: 2.6M

Model size: 9.9 MB

[Dataset] Loaded 146871 samples for train

REAL: 20970, FAKE: 125901

[Dataset] Loaded 31487 samples for val

REAL: 4500, FAKE: 26987

⚡ AMP (fp16) enabled – expect 2-3× speedup

```
=====
Training mobilenetv2_baseline
Epochs: 1 → 50, Device: cuda
=====
```

Epoch [1/50] (3280.0s)

Train – Loss: 0.2154, Acc: 0.9121, AUC: 0.9299

Val – Loss: 0.5000, Acc: 0.8201, AUC: 0.9111

Group Acc Gap: 0.2025 (min: 0.7525, max: 0.9550)

Female-Asian : 0.8234

Female-Black : 0.9463

Female-Other : 0.8142

Female-White : 0.8310

Male-Asian : 0.7629

Male-Black : 0.9550

Male-Other : 0.7525

Male-White : 0.8676

★ New best AUC: 0.9111 – saved to D:\M3 Projects\DeepFake_Research\outputs\models\mobilenetv2_baseline_best.pth

Epoch [2/50] (2269.7s)

Train – Loss: 0.1120, Acc: 0.9551, AUC: 0.9834

Val – Loss: 0.2610, Acc: 0.9051, AUC: 0.9419

Group Acc Gap: 0.0844 (min: 0.8685, max: 0.9529)

Female-Asian : 0.9260

Female-Black : 0.9529

Female-Other : 0.9073

Female-White : 0.9162

Male-Asian : 0.8836

Male-Black : 0.9452

Male-Other : 0.8685

Male-White : 0.9086

★ New best AUC: 0.9419 – saved to D:\M3 Projects\DeepFake_Research\outputs\models\mobilenetv2_baseline_best.pth

Epoch [3/50] (1698.1s)

Train — Loss: 0.0809, Acc: 0.9682, AUC: 0.9915

Val — Loss: 0.2895, Acc: 0.9042, AUC: 0.9522

Group Acc Gap: 0.0955 (min: 0.8738, max: 0.9693)

Female-Asian : 0.9390

Female-Black : 0.9490

Female-Other : 0.9047

Female-White : 0.9035

Male-Asian : 0.8869

Male-Black : 0.9693

Male-Other : 0.8738

Male-White : 0.9114

★ New best AUC: 0.9522 — saved to D:\M3 Projects\DeepFake_Research\outputs\models\mobilenetv2_baseline_best.pth

Epoch [4/50] (2151.4s)

Train — Loss: 0.0636, Acc: 0.9757, AUC: 0.9947

Val — Loss: 0.2991, Acc: 0.9072, AUC: 0.9527

Group Acc Gap: 0.0982 (min: 0.8766, max: 0.9748)

Female-Asian : 0.9429

Female-Black : 0.9581

Female-Other : 0.9069

Female-White : 0.9073

Male-Asian : 0.9176

Male-Black : 0.9748

Male-Other : 0.8766

Male-White : 0.9000

★ New best AUC: 0.9527 — saved to D:\M3 Projects\DeepFake_Research\outputs\models\mobilenetv2_baseline_best.pth

Epoch [5/50] (1707.1s)

Train — Loss: 0.0543, Acc: 0.9796, AUC: 0.9962

Val — Loss: 0.3218, Acc: 0.9055, AUC: 0.9479

Group Acc Gap: 0.0898 (min: 0.8751, max: 0.9649)

Female-Asian : 0.9362

Female-Black : 0.9424

Female-Other : 0.9029

Female-White : 0.9063

Male-Asian : 0.9186

Male-Black : 0.9649

Male-Other : 0.8751

Male-White : 0.9051

Epoch [6/50] (1919.6s)

Train — Loss: 0.0460, Acc: 0.9824, AUC: 0.9973

Val — Loss: 0.3055, Acc: 0.9242, AUC: 0.9544

Group Acc Gap: 0.0655 (min: 0.9104, max: 0.9759)

Female-Asian : 0.9415

Female-Black : 0.9332

Female-Other : 0.9251

Female-White : 0.9239

Male-Asian : 0.9321

Male-Black : 0.9759

Male-Other : 0.9104

Male-White : 0.9167

★ New best AUC: 0.9544 — saved to D:\M3 Projects\DeepFake_Research\outputs\models
\mobilenetv2_baseline_best.pth

Epoch [7/50] (2712.5s)

Train — Loss: 0.0414, Acc: 0.9848, AUC: 0.9978

Val — Loss: 0.3382, Acc: 0.9088, AUC: 0.9476

Group Acc Gap: 0.0952 (min: 0.8818, max: 0.9770)

Female-Asian : 0.9214

Female-Black : 0.9568

Female-Other : 0.9183

Female-White : 0.9076

Male-Asian : 0.9149

Male-Black : 0.9770

Male-Other : 0.8818

Male-White : 0.9062

Epoch [8/50] (1217.2s)

Train — Loss: 0.0380, Acc: 0.9858, AUC: 0.9981

Val — Loss: 0.3622, Acc: 0.9138, AUC: 0.9471

Group Acc Gap: 0.0851 (min: 0.8929, max: 0.9781)

Female-Asian : 0.9263

Female-Black : 0.9175

Female-Other : 0.9220

Female-White : 0.9205

Male-Asian : 0.9143

Male-Black : 0.9781

Male-Other : 0.8929

Male-White : 0.8983

Epoch [9/50] (1208.9s)

Train — Loss: 0.0354, Acc: 0.9868, AUC: 0.9984

Val — Loss: 0.3536, Acc: 0.9180, AUC: 0.9517

Group Acc Gap: 0.0748 (min: 0.8923, max: 0.9671)

Female-Asian	: 0.9457
Female-Black	: 0.9215
Female-Other	: 0.9240
Female-White	: 0.9222
Male-Asian	: 0.9418
Male-Black	: 0.9671
Male-Other	: 0.8923
Male-White	: 0.9005

Epoch [10/50] (1214.2s)

Train — Loss: 0.0315, Acc: 0.9881, AUC: 0.9987

Val — Loss: 0.3770, Acc: 0.9214, AUC: 0.9503

Group Acc Gap: 0.0781 (min: 0.8989, max: 0.9770)

Female-Asian	: 0.9404
Female-Black	: 0.9319
Female-Other	: 0.9187
Female-White	: 0.9304
Male-Asian	: 0.9407
Male-Black	: 0.9770
Male-Other	: 0.8989
Male-White	: 0.9027

Epoch [11/50] (1205.4s)

Train — Loss: 0.0305, Acc: 0.9888, AUC: 0.9988

Val — Loss: 0.3826, Acc: 0.9141, AUC: 0.9443

Group Acc Gap: 0.0574 (min: 0.8988, max: 0.9561)

Female-Asian	: 0.9228
Female-Black	: 0.9188
Female-Other	: 0.9128
Female-White	: 0.9184
Male-Asian	: 0.9251
Male-Black	: 0.9561
Male-Other	: 0.8988
Male-White	: 0.9079

Epoch [12/50] (1223.8s)

Train — Loss: 0.0275, Acc: 0.9897, AUC: 0.9990

Val — Loss: 0.4232, Acc: 0.9122, AUC: 0.9435

Group Acc Gap: 0.0686 (min: 0.8919, max: 0.9605)

Female-Asian	: 0.9253
Female-Black	: 0.8992
Female-Other	: 0.9148
Female-White	: 0.9237
Male-Asian	: 0.9186
Male-Black	: 0.9605
Male-Other	: 0.8937
Male-White	: 0.8919

Epoch [13/50] (2155.5s)

Train — Loss: 0.0257, Acc: 0.9903, AUC: 0.9992

Val — Loss: 0.3424, Acc: 0.9168, AUC: 0.9517

Group Acc Gap: 0.0877 (min: 0.8848, max: 0.9726)

Female-Asian : 0.9337

Female-Black : 0.9306

Female-Other : 0.9176

Female-White : 0.9292

Male-Asian : 0.9267

Male-Black : 0.9726

Male-Other : 0.8848

Male-White : 0.9042

Epoch [14/50] (1317.0s)

Train — Loss: 0.0243, Acc: 0.9911, AUC: 0.9992

Val — Loss: 0.4773, Acc: 0.9026, AUC: 0.9425

Group Acc Gap: 0.0864 (min: 0.8620, max: 0.9485)

Female-Asian : 0.9295

Female-Black : 0.8796

Female-Other : 0.9185

Female-White : 0.9123

Male-Asian : 0.9089

Male-Black : 0.9485

Male-Other : 0.8620

Male-White : 0.8959

Epoch [15/50] (1207.2s)

Train — Loss: 0.0231, Acc: 0.9911, AUC: 0.9993

Val — Loss: 0.4074, Acc: 0.9081, AUC: 0.9469

Group Acc Gap: 0.0806 (min: 0.8789, max: 0.9594)

Female-Asian : 0.9341

Female-Black : 0.8796

Female-Other : 0.9172

Female-White : 0.9133

Male-Asian : 0.9127

Male-Black : 0.9594

Male-Other : 0.8789

Male-White : 0.9036

```
Epoch [16/50] (1207.5s)
  Train — Loss: 0.0211, Acc: 0.9921, AUC: 0.9994
  Val   — Loss: 0.4744, Acc: 0.9170, AUC: 0.9479
  Group Acc Gap: 0.0631 (min: 0.8952, max: 0.9583)
    Female-Asian      : 0.9387
    Female-Black      : 0.9110
    Female-Other      : 0.9172
    Female-White      : 0.9228
    Male-Asian        : 0.9397
    Male-Black        : 0.9583
    Male-Other        : 0.8952
    Male-White        : 0.9038
```

Early stopping at epoch 16 (patience=10)

Training complete. Best AUC: 0.9544

✅ Baseline training done. Best AUC: 0.9544

Cell 5 — Train Fair Student (Fairness-Aware Distillation) — OUR METHOD

Core contribution. Skips if checkpoint exists.

```
In [6]: from src.training.train_distill import train_fair_distillation

FAIR_WEIGHTS = os.path.join(MODELS_DIR, 'fair_student_best_auc.pth')
FAIR_EPOCHS   = 50    # ← change as needed
FAIR_BATCH    = 32    # ← distillation: teacher+student both in VRAM
FAIR_ALPHA    = 0.7    # distillation loss weight
FAIR_BETA     = 0.2    # fairness loss weight ← our novel term
FAIR_GAMMA    = 0.1    # classification loss weight

if os.path.exists(FAIR_WEIGHTS):
    print(f'✅ Fair student checkpoint found — skipping training.')
    ckpt = torch.load(FAIR_WEIGHTS, map_location=DEVICE, weights_only=False)
    print(f'    Saved at epoch {ckpt["epoch"]+1}, val AUC = {ckpt["val_auc"]:.4f}')
    print(f'    Val metrics: {ckpt.get("val_metrics", {})}')
else:
    print(f'🔧 Training Fair Student (Fairness-Aware Distillation)...')
    print(f'    Loss: {FAIR_ALPHA}·L_distill + {FAIR_BETA}·L_fairness + {FAIR_GAMMA}·L_cls

    # Load teacher
    teacher = build_teacher(pretrained=True, device=DEVICE)
    ckpt = torch.load(TEACHER_WEIGHTS, map_location=DEVICE, weights_only=False)
    teacher.load_state_dict(ckpt['model_state_dict'])
    print(f'    Teacher loaded (val AUC={ckpt["val_auc"]:.4f})')

    # Fresh student
    fair_student = build_student(pretrained=True, device=DEVICE)

    loaders = create_dataloaders(
        os.path.join(SPLITS_DIR, 'train.csv'),
```

```

    os.path.join(SPLITS_DIR, 'val.csv'),
    batch_size=FAIR_BATCH,
    fair_sampling=True
)
optimizer = torch.optim.AdamW(fair_student.parameters(), lr=1e-4, weight_decay=
scheduler = torch.optim.lr_scheduler.ReduceLROnPlateau(
    optimizer, mode='max', factor=0.5, patience=5, verbose=True
)
torch.cuda.empty_cache()
torch.backends.cudnn.benchmark = False
torch.backends.cudnn.deterministic = True
results = train_fair_distillation(
    fair_student, teacher,
    loaders['train'], loaders['val'],
    optimizer, scheduler,
    num_epochs=FAIR_EPOCHS,
    alpha=FAIR_ALPHA, beta=FAIR_BETA, gamma=FAIR_GAMMA,
    temperature=DISTILL_TEMPERATURE,
    model_name='fair_student',
    device=DEVICE
)
print(f'\n✅ Fair distillation done.')
print(f'    Best AUC: {results["best_auc"]:.4f}')
print(f'    Best Fairness Gap: {results["best_fairness_gap"]:.4f}')

```

🚀 Training Fair Student (Fairness-Aware Distillation)...

Loss: $0.7 \cdot L_{\text{distill}} + 0.2 \cdot L_{\text{fairness}} + 0.1 \cdot L_{\text{cls}}$

[Teacher] XceptionNet loaded:

Total params: 21.9M

Trainable params: 21.9M

Model size: 83.6 MB

Teacher loaded (val AUC=0.9671)

[Student] MobileNetV2 loaded:

Total params: 2.6M

Trainable params: 2.6M

Model size: 9.9 MB

[Dataset] Loaded 146871 samples for train

REAL: 20970, FAKE: 125901

[Dataset] Loaded 31487 samples for val

REAL: 4500, FAKE: 26987

=====

Fairness-Aware Knowledge Distillation

Student: fair_student

Loss: $\alpha=0.7 \cdot L_d + \beta=0.2 \cdot L_f + \gamma=0.1 \cdot L_c$

Temperature: 4.0, Epochs: 50

AMP (fp16): ✅ ON

=====

Epoch [1/50] (1333.8s) $\beta=0.2000$
Train – Loss: 1.7215 (Ld:2.395 Lf:0.095 Lc:0.262)
Train – Acc: 0.9174, AUC: 0.9389
Val – Acc: 0.9034, AUC: 0.9453, AccGap: 0.0656
Female-Asian : 0.9253
Female-Black : 0.9241
Female-Other : 0.9034
Female-White : 0.8935
Male-Asian : 0.9073
Male-Black : 0.9572
Male-Other : 0.8916
Male-White : 0.9114
★ Best AUC: 0.9453 – saved
★ Best Fairness (gap=0.0656) – saved

Epoch [2/50] (1346.9s) $\beta=0.2000$
Train – Loss: 0.9276 (Ld:1.286 Lf:0.066 Lc:0.146)
Train – Acc: 0.9537, AUC: 0.9811
Val – Acc: 0.9168, AUC: 0.9472, AccGap: 0.0756
Female-Asian : 0.9383
Female-Black : 0.9555
Female-Other : 0.9064
Female-White : 0.9036
Male-Asian : 0.9127
Male-Black : 0.9792
Male-Other : 0.9174
Male-White : 0.9244
★ Best AUC: 0.9472 – saved

Epoch [3/50] (1345.4s) $\beta=0.2000$
Train – Loss: 0.7343 (Ld:1.017 Lf:0.055 Lc:0.118)
Train – Acc: 0.9637, AUC: 0.9875
Val – Acc: 0.9190, AUC: 0.9491, AccGap: 0.0484
Female-Asian : 0.9295
Female-Black : 0.9542
Female-Other : 0.9148
Female-White : 0.9258
Male-Asian : 0.9154
Male-Black : 0.9550
Male-Other : 0.9073
Male-White : 0.9066
★ Best AUC: 0.9491 – saved
★ Best Fairness (gap=0.0484) – saved

Epoch [4/50] (1343.1s) $\beta=0.2000$
Train – Loss: 0.6210 (Ld:0.859 Lf:0.048 Lc:0.101)
Train – Acc: 0.9693, AUC: 0.9904
Val – Acc: 0.9235, AUC: 0.9514, AccGap: 0.0784
Female-Asian : 0.9485
Female-Black : 0.9437
Female-Other : 0.9163
Female-White : 0.9207
Male-Asian : 0.9224
Male-Black : 0.9923
Male-Other : 0.9140
Male-White : 0.9176
★ Best AUC: 0.9514 – saved

Epoch [5/50] (1340.6s) $\beta=0.2000$
Train – Loss: 0.5606 (Ld:0.775 Lf:0.045 Lc:0.090)
Train – Acc: 0.9725, AUC: 0.9920
Val – Acc: 0.9161, AUC: 0.9504, AccGap: 0.0795
Female-Asian : 0.9475
Female-Black : 0.9175
Female-Other : 0.9093
Female-White : 0.9133
Male-Asian : 0.9041
Male-Black : 0.9836
Male-Other : 0.9096
Male-White : 0.9095

Epoch [6/50] (1338.7s) $\beta=0.2000$
Train – Loss: 0.5078 (Ld:0.701 Lf:0.042 Lc:0.084)
Train – Acc: 0.9747, AUC: 0.9929
Val – Acc: 0.9236, AUC: 0.9565, AccGap: 0.0588
Female-Asian : 0.9411
Female-Black : 0.9581
Female-Other : 0.9244
Female-White : 0.9184
Male-Asian : 0.9127
Male-Black : 0.9715
Male-Other : 0.9229
Male-White : 0.9128
★ Best AUC: 0.9565 – saved

Epoch [7/50] (1368.0s) $\beta=0.2000$
Train — Loss: 0.4732 (Ld:0.653 Lf:0.040 Lc:0.079)
Train — Acc: 0.9762, AUC: 0.9937
Val — Acc: 0.9210, AUC: 0.9443, AccGap: 0.0694
Female-Asian : 0.9387
Female-Black : 0.9450
Female-Other : 0.9260
Female-White : 0.9271
Male-Asian : 0.9203
Male-Black : 0.9704
Male-Other : 0.9010
Male-White : 0.9051

Epoch [8/50] (1347.7s) $\beta=0.2000$
Train — Loss: 0.4411 (Ld:0.608 Lf:0.039 Lc:0.074)
Train — Acc: 0.9775, AUC: 0.9944
Val — Acc: 0.9234, AUC: 0.9430, AccGap: 0.0736
Female-Asian : 0.9274
Female-Black : 0.9542
Female-Other : 0.9317
Female-White : 0.9274
Male-Asian : 0.9213
Male-Black : 0.9759
Male-Other : 0.9136
Male-White : 0.9022

Epoch [9/50] (1349.0s) $\beta=0.2000$
Train — Loss: 0.4277 (Ld:0.590 Lf:0.037 Lc:0.072)
Train — Acc: 0.9788, AUC: 0.9945
Val — Acc: 0.9263, AUC: 0.9517, AccGap: 0.0739
Female-Asian : 0.9306
Female-Black : 0.9372
Female-Other : 0.9308
Female-White : 0.9309
Male-Asian : 0.9348
Male-Black : 0.9803
Male-Other : 0.9167
Male-White : 0.9064

Epoch [10/50] (1342.8s) $\beta=0.2000$
Train — Loss: 0.4005 (Ld:0.553 Lf:0.035 Lc:0.065)
Train — Acc: 0.9802, AUC: 0.9953
Val — Acc: 0.9255, AUC: 0.9521, AccGap: 0.0771
Female-Asian : 0.9468
Female-Black : 0.9280
Female-Other : 0.9315
Female-White : 0.9325
Male-Asian : 0.9380
Male-Black : 0.9803
Male-Other : 0.9031
Male-White : 0.9051

```
Epoch [11/50] (1348.0s)  $\beta=0.2000$   
  Train — Loss: 0.3869 (Ld:0.534 Lf:0.035 Lc:0.064)  
  Train — Acc: 0.9803, AUC: 0.9955  
  Val   — Acc: 0.9185, AUC: 0.9511, AccGap: 0.0902  
    Female-Asian      : 0.9330  
    Female-Black      : 0.9346  
    Female-Other      : 0.9240  
    Female-White      : 0.9175  
    Male-Asian        : 0.9310  
    Male-Black        : 0.9901  
    Male-Other        : 0.8999  
    Male-White        : 0.9095
```

```
Epoch [12/50] (1349.8s)  $\beta=0.2000$   
  Train — Loss: 0.3685 (Ld:0.508 Lf:0.034 Lc:0.062)  
  Train — Acc: 0.9812, AUC: 0.9956  
  Val   — Acc: 0.9218, AUC: 0.9426, AccGap: 0.0929  
    Female-Asian      : 0.9299  
    Female-Black      : 0.8940  
    Female-Other      : 0.9310  
    Female-White      : 0.9232  
    Male-Asian        : 0.9407  
    Male-Black        : 0.9868  
    Male-Other        : 0.9075  
    Male-White        : 0.9082
```

```
Epoch [13/50] (1349.4s)  $\beta=0.2000$   
  Train — Loss: 0.2893 (Ld:0.398 Lf:0.029 Lc:0.049)  
  Train — Acc: 0.9850, AUC: 0.9969  
  Val   — Acc: 0.9254, AUC: 0.9488, AccGap: 0.0467  
    Female-Asian      : 0.9489  
    Female-Black      : 0.9359  
    Female-Other      : 0.9313  
    Female-White      : 0.9217  
    Male-Asian        : 0.9332  
    Male-Black        : 0.9616  
    Male-Other        : 0.9149  
    Male-White        : 0.9152  
★ Best Fairness (gap=0.0467) — saved
```



```
Epoch [14/50] (1349.3s)  $\beta=0.2000$ 
  Train — Loss: 0.2696 (Ld:0.371 Lf:0.028 Lc:0.046)
  Train — Acc: 0.9860, AUC: 0.9971
  Val   — Acc: 0.9219, AUC: 0.9403, AccGap: 0.0589
    Female-Asian      : 0.9263
    Female-Black      : 0.9188
    Female-Other      : 0.9299
    Female-White      : 0.9249
    Male-Asian        : 0.9402
    Male-Black        : 0.9638
    Male-Other        : 0.9106
    Male-White        : 0.9049
```

```
Epoch [15/50] (1349.3s)  $\beta=0.2000$ 
  Train — Loss: 0.2607 (Ld:0.358 Lf:0.027 Lc:0.044)
  Train — Acc: 0.9863, AUC: 0.9973
  Val   — Acc: 0.9182, AUC: 0.9407, AccGap: 0.0707
    Female-Asian      : 0.9232
    Female-Black      : 0.9215
    Female-Other      : 0.9240
    Female-White      : 0.9222
    Male-Asian        : 0.9348
    Male-Black        : 0.9660
    Male-Other        : 0.8954
    Male-White        : 0.9147
```

```
Epoch [16/50] (1345.1s)  $\beta=0.2000$ 
  Train — Loss: 0.2475 (Ld:0.340 Lf:0.027 Lc:0.044)
  Train — Acc: 0.9864, AUC: 0.9975
  Val   — Acc: 0.9205, AUC: 0.9389, AccGap: 0.0620
    Female-Asian      : 0.9151
    Female-Black      : 0.9123
    Female-Other      : 0.9247
    Female-White      : 0.9238
    Male-Asian        : 0.9327
    Male-Black        : 0.9715
    Male-Other        : 0.9128
    Male-White        : 0.9095
```

Early stopping at epoch 16

Fair distillation complete.

Best AUC: 0.9565

Best Fairness Gap: 0.0467

✅ Fair distillation done.

Best AUC: 0.9565

Best Fairness Gap: 0.0467



Cell 6 — Train Standard Distillation (Ablation — $\beta=0$, no fairness loss)

Needed for the ablation table. Skips if checkpoint exists.

```
In [7]: STD_WEIGHTS = os.path.join(MODELS_DIR, 'std_distill_student_best_auc.pth')
STD_EPOCHS = 50
STD_BATCH = 32      # ← distillation: teacher+student both in VRAM

if os.path.exists(STD_WEIGHTS):
    print(f'✅ Standard distillation checkpoint found – skipping.')
    ckpt = torch.load(STD_WEIGHTS, map_location=DEVICE, weights_only=False)
    print(f'    Saved at epoch {ckpt["epoch"]+1}, val AUC = {ckpt["val_auc"]:.4f}')
else:
    print(f'🚀 Training Standard Distillation student ( $\beta=0$ , no fairness loss)...')

    teacher = build_teacher(pretrained=True, device=DEVICE)
    ckpt = torch.load(TEACHER_WEIGHTS, map_location=DEVICE, weights_only=False)
    teacher.load_state_dict(ckpt['model_state_dict'])

    std_student = build_student(pretrained=True, device=DEVICE)
    loaders = create_dataloaders(
        os.path.join(SPLITS_DIR, 'train.csv'),
        os.path.join(SPLITS_DIR, 'val.csv'),
        batch_size=STD_BATCH,
        fair_sampling=True
    )
    optimizer = torch.optim.AdamW(std_student.parameters(), lr=1e-4, weight_decay=1)
    scheduler = torch.optim.lr_scheduler.ReduceLROnPlateau(
        optimizer, mode='max', factor=0.5, patience=5, verbose=True
    )
    torch.cuda.empty_cache()
    results = train_fair_distillation(
        std_student, teacher,
        loaders['train'], loaders['val'],
        optimizer, scheduler,
        num_epochs=STD_EPOCHS,
        alpha=0.9, beta=0.0, gamma=0.1,      # ←  $\beta=0$ : no fairness constraint
        temperature=DISTILL_TEMPERATURE,
        model_name='std_distill_student',
        device=DEVICE
    )
    print(f'\n✅ Standard distillation done. Best AUC: {results["best_auc"]:.4f}')
```

🚀 Training Standard Distillation student ($\beta=0$, no fairness loss)...

[Teacher] XceptionNet loaded:

Total params: 21.9M
Trainable params: 21.9M
Model size: 83.6 MB

[Student] MobileNetV2 loaded:

Total params: 2.6M
Trainable params: 2.6M
Model size: 9.9 MB

[Dataset] Loaded 146871 samples for train

REAL: 20970, FAKE: 125901

[Dataset] Loaded 31487 samples for val

REAL: 4500, FAKE: 26987

=====

Fairness-Aware Knowledge Distillation

Student: std_distill_student

Loss: $\alpha=0.9 \cdot L_d + \beta=0.0 \cdot L_f + \gamma=0.1 \cdot L_c$

Temperature: 4.0, Epochs: 50

AMP (fp16): ☒ ON

=====

Epoch [1/50] (1337.0s) $\beta=0.0000$

Train - Loss: 2.1996 (Ld:2.415 Lf:0.095 Lc:0.263)

Train - Acc: 0.9177, AUC: 0.9384

Val - Acc: 0.8897, AUC: 0.9394, AccGap: 0.1120

Female-Asian	: 0.9225
Female-Black	: 0.9503
Female-Other	: 0.8926
Female-White	: 0.8971
Male-Asian	: 0.8572
Male-Black	: 0.9616
Male-Other	: 0.8496
Male-White	: 0.8937

★ Best AUC: 0.9394 - saved

★ Best Fairness (gap=0.1120) - saved

Epoch [2/50] (1339.3s) $\beta=0.0000$

Train - Loss: 1.1861 (Ld:1.301 Lf:0.065 Lc:0.150)

Train - Acc: 0.9530, AUC: 0.9803

Val - Acc: 0.9031, AUC: 0.9408, AccGap: 0.1365

Female-Asian	: 0.9228
Female-Black	: 0.9149
Female-Other	: 0.9034
Female-White	: 0.9154
Male-Asian	: 0.8448
Male-Black	: 0.9814
Male-Other	: 0.8742
Male-White	: 0.9095

★ Best AUC: 0.9408 - saved

Epoch [3/50] (1343.2s) $\beta=0.0000$
Train – Loss: 0.9259 (Ld:1.015 Lf:0.056 Lc:0.120)
Train – Acc: 0.9628, AUC: 0.9873
Val – Acc: 0.9161, AUC: 0.9506, AccGap: 0.0936
Female-Asian : 0.9299
Female-Black : 0.9319
Female-Other : 0.9227
Female-White : 0.9155
Male-Asian : 0.8955
Male-Black : 0.9890
Male-Other : 0.8971
Male-White : 0.9193
★ Best AUC: 0.9506 – saved
★ Best Fairness (gap=0.0936) – saved

Epoch [4/50] (1339.1s) $\beta=0.0000$
Train – Loss: 0.7866 (Ld:0.863 Lf:0.049 Lc:0.103)
Train – Acc: 0.9692, AUC: 0.9903
Val – Acc: 0.9085, AUC: 0.9431, AccGap: 0.1036
Female-Asian : 0.9186
Female-Black : 0.9686
Female-Other : 0.9148
Female-White : 0.9085
Male-Asian : 0.8766
Male-Black : 0.9803
Male-Other : 0.8918
Male-White : 0.9071

Epoch [5/50] (1338.2s) $\beta=0.0000$
Train – Loss: 0.7012 (Ld:0.769 Lf:0.046 Lc:0.092)
Train – Acc: 0.9718, AUC: 0.9917
Val – Acc: 0.9190, AUC: 0.9410, AccGap: 0.0673
Female-Asian : 0.9284
Female-Black : 0.9188
Female-Other : 0.9317
Female-White : 0.9235
Male-Asian : 0.9116
Male-Black : 0.9671
Male-Other : 0.9072
Male-White : 0.8998
★ Best Fairness (gap=0.0673) – saved

Epoch [6/50] (2600.0s) $\beta=0.0000$
Train – Loss: 0.6523 (Ld:0.715 Lf:0.043 Lc:0.086)
Train – Acc: 0.9741, AUC: 0.9928
Val – Acc: 0.9207, AUC: 0.9482, AccGap: 0.0570
Female-Asian : 0.9330
Female-Black : 0.9149
Female-Other : 0.9328
Female-White : 0.9304
Male-Asian : 0.9095
Male-Black : 0.9550
Male-Other : 0.9065
Male-White : 0.8981
★ Best Fairness (gap=0.0570) – saved

Epoch [7/50] (1578.3s) $\beta=0.0000$
Train – Loss: 0.6011 (Ld:0.659 Lf:0.040 Lc:0.080)
Train – Acc: 0.9758, AUC: 0.9935
Val – Acc: 0.9190, AUC: 0.9439, AccGap: 0.0835
Female-Asian : 0.9454
Female-Black : 0.9437
Female-Other : 0.9190
Female-White : 0.9214
Male-Asian : 0.9208
Male-Black : 0.9748
Male-Other : 0.9117
Male-White : 0.8913

Epoch [8/50] (1324.8s) $\beta=0.0000$
Train – Loss: 0.5695 (Ld:0.625 Lf:0.039 Lc:0.075)
Train – Acc: 0.9774, AUC: 0.9941
Val – Acc: 0.9222, AUC: 0.9448, AccGap: 0.0571
Female-Asian : 0.9390
Female-Black : 0.9529
Female-Other : 0.9163
Female-White : 0.9272
Male-Asian : 0.9089
Male-Black : 0.9660
Male-Other : 0.9098
Male-White : 0.9152

Epoch [9/50] (1338.4s) $\beta=0.0000$
Train – Loss: 0.5370 (Ld:0.589 Lf:0.038 Lc:0.072)
Train – Acc: 0.9782, AUC: 0.9946
Val – Acc: 0.9128, AUC: 0.9464, AccGap: 0.0925
Female-Asian : 0.9344
Female-Black : 0.8979
Female-Other : 0.9181
Female-White : 0.9159
Male-Asian : 0.9122
Male-Black : 0.9857
Male-Other : 0.8933
Male-White : 0.9022

Epoch [10/50] (1338.7s) $\beta=0.0000$
Train – Loss: 0.4127 (Ld:0.452 Lf:0.032 Lc:0.056)
Train – Acc: 0.9827, AUC: 0.9962
Val – Acc: 0.9230, AUC: 0.9503, AccGap: 0.0618
Female-Asian : 0.9415
Female-Black : 0.9319
Female-Other : 0.9319
Female-White : 0.9266
Male-Asian : 0.9213
Male-Black : 0.9649
Male-Other : 0.9031
Male-White : 0.9130

Epoch [11/50] (1345.8s) $\beta=0.0000$
Train – Loss: 0.3794 (Ld:0.416 Lf:0.030 Lc:0.052)
Train – Acc: 0.9842, AUC: 0.9967
Val – Acc: 0.9235, AUC: 0.9474, AccGap: 0.0706
Female-Asian : 0.9351
Female-Black : 0.9372
Female-Other : 0.9293
Female-White : 0.9238
Male-Asian : 0.9186
Male-Black : 0.9770
Male-Other : 0.9180
Male-White : 0.9064

Epoch [12/50] (1333.3s) $\beta=0.0000$
Train – Loss: 0.3562 (Ld:0.390 Lf:0.030 Lc:0.050)
Train – Acc: 0.9845, AUC: 0.9969
Val – Acc: 0.9182, AUC: 0.9342, AccGap: 0.0682
Female-Asian : 0.9323
Female-Black : 0.9097
Female-Other : 0.9286
Female-White : 0.9237
Male-Asian : 0.9057
Male-Black : 0.9649
Male-Other : 0.9089
Male-White : 0.8968

```
Epoch [13/50] (1335.1s)  $\beta=0.0000$ 
  Train — Loss: 0.3543 (Ld:0.388 Lf:0.029 Lc:0.049)
  Train — Acc: 0.9848, AUC: 0.9970
  Val   — Acc: 0.9256, AUC: 0.9447, AccGap: 0.0673
    Female-Asian      : 0.9510
    Female-Black      : 0.9385
    Female-Other      : 0.9247
    Female-White      : 0.9277
    Male-Asian        : 0.9316
    Male-Black        : 0.9759
    Male-Other        : 0.9130
    Male-White        : 0.9086
```

Early stopping at epoch 13

Fair distillation complete.

Best AUC: 0.9506

Best Fairness Gap: 0.0570

✅ Standard distillation done. Best AUC: 0.9506



Cell 7 — Load All Trained Models for Evaluation

```
In [8]: def load_model(model, weights_path, device):
        if not os.path.exists(weights_path):
            print(f'[WARN] Not found: {weights_path}')
            return None
        ckpt = torch.load(weights_path, map_location=device, weights_only=False)
        model.load_state_dict(ckpt['model_state_dict'])
        model.eval()
        return model

# Teacher
teacher_model = build_teacher(pretrained=True, device=DEVICE)
teacher_model = load_model(teacher_model, os.path.join(MODELS_DIR, 'xception_teacher_0.9506.pth'))

# Baseline student
baseline_model = build_student(pretrained=True, device=DEVICE)
baseline_model = load_model(baseline_model, os.path.join(MODELS_DIR, 'mobilenetv2_baseline_0.9256.pth'))

# Standard distillation (ablation)
std_model = build_student(pretrained=True, device=DEVICE)
std_model = load_model(std_model, os.path.join(MODELS_DIR, 'std_distill_student_best_0.9447.pth'))

# Fair distillation (ours)
fair_model = build_student(pretrained=True, device=DEVICE)
fair_model = load_model(fair_model, os.path.join(MODELS_DIR, 'fair_student_best_auc_0.9506.pth'))

models = {
    'XceptionNet (Teacher)': teacher_model,
    'MobileNetV2 (Baseline)': baseline_model,
    'MobileNetV2 (Std Distill)': std_model,
    'MobileNetV2 (Fair Distill)': fair_model,
```

```

}
print('Models loaded:', [k for k,v in models.items() if v is not None])

```

[Teacher] XceptionNet loaded:
 Total params: 21.9M
 Trainable params: 21.9M
 Model size: 83.6 MB

[Student] MobileNetV2 loaded:
 Total params: 2.6M
 Trainable params: 2.6M
 Model size: 9.9 MB

[Student] MobileNetV2 loaded:
 Total params: 2.6M
 Trainable params: 2.6M
 Model size: 9.9 MB

[Student] MobileNetV2 loaded:
 Total params: 2.6M
 Trainable params: 2.6M
 Model size: 9.9 MB

Models loaded: ['XceptionNet (Teacher)', 'MobileNetV2 (Baseline)', 'MobileNetV2 (Std Distill)', 'MobileNetV2 (Fair Distill)']



Cell 8 — In-Domain Evaluation (FF+ + Test Set)

```

In [9]: test_dataset = DeepfakeDataset(os.path.join(SPLITS_DIR, 'test.csv'), split='test')
test_loader = DataLoader(test_dataset, batch_size=BATCH_SIZE, shuffle=False, num_w

all_results = {}
for name, model in models.items():
    if model is None:
        print(f'[SKIP] {name} - no checkpoint')
        continue
    r = evaluate_model(model, test_loader, DEVICE, name)
    save_results(r, RESULTS_DIR, filename=name.replace(' ', '_').replace('(', '').rep
    all_results[name] = r

print('\n✅ In-domain evaluation complete')

```

[Dataset] Loaded 31474 samples for test
 REAL: 4527, FAKE: 26947

Evaluating XceptionNet (Teacher): 100% |██████████| 984/984 [06:54<00:00, 2.37it/s]


```
=====
EVALUATION: XceptionNet (Teacher)
=====
Samples:      31474
Overall Acc:  0.9173
Overall AUC:  0.9351
Inference:    0.2 ms/image
Model Size:   83.4 MB (21.9M params)

=====
FAIRNESS REPORT: XceptionNet (Teacher)
=====

Group              Acc      AUC      FPR      TPR      PPV      Count
-----
Female-Asian       0.923   0.945   0.410   0.976   0.937    2931
Female-Black       0.920   0.960   0.402   0.974   0.935     778
Female-Other       0.930   0.951   0.315   0.978   0.941   4563
Female-White       0.932   0.944   0.349   0.978   0.945   9631
Male-Asian         0.887   0.914   0.401   0.963   0.900   1699
Male-Black         0.905   0.810   0.952   0.962   0.939   1015
Male-Other         0.909   0.933   0.400   0.956   0.940   6109
Male-White         0.895   0.914   0.569   0.972   0.911   4748

Fairness Metric              Value      Target      Pass
-----
FFPR Gap (FPR)              0.6373      0.12         X
FEO Gap (Equalized Odds)    0.6373      0.10         X
FDP Gap (Demographic Parity) 0.1164      0.10         X
FOAE Gap (Accuracy)         0.0453      0.08         ✓
AUC Gap                     0.1504      0.05         X

Worst group (accuracy): Male-Asian
Best group (accuracy):  Female-White
=====

Results saved to: D:\M3 Projects\DeepFake_Research\outputs\results\XceptionNet_Teacher.csv
Evaluating MobileNetV2 (Baseline): 100%|██████████| 984/984 [01:26<00:00, 11.36it/s]
```

```
=====
EVALUATION: MobileNetV2 (Baseline)
=====

Samples:      31474
Overall Acc:   0.9193
Overall AUC:   0.9521
Inference:     0.2 ms/image
Model Size:    9.7 MB (2.6M params)

=====

FAIRNESS REPORT: MobileNetV2 (Baseline)
=====

Group              Acc      AUC      FPR      TPR      PPV      Count
-----
Female-Asian       0.936   0.971   0.296   0.973   0.954   2931
Female-Black       0.955   0.988   0.277   0.994   0.955    778
Female-Other       0.933   0.967   0.278   0.974   0.947   4563
Female-White       0.926   0.956   0.298   0.963   0.952   9631
Male-Asian         0.889   0.954   0.471   0.984   0.887   1699
Male-Black         0.928   0.927   0.794   0.976   0.949   1015
Male-Other         0.900   0.933   0.451   0.954   0.932   6109
Male-White         0.909   0.937   0.445   0.968   0.929   4748

Fairness Metric              Value      Target      Pass
-----
FFPR Gap (FPR)              0.5169      0.12         X
FEO Gap (Equalized Odds)     0.5169      0.10         X
FDP Gap (Demographic Parity) 0.1046      0.10         X
FOAE Gap (Accuracy)          0.0663      0.08         ✓
AUC Gap                      0.0607      0.05         X

Worst group (accuracy): Male-Asian
Best group (accuracy):  Female-Black
=====

Results saved to: D:\M3 Projects\DeepFake_Research\outputs\results\MobileNetV2_Baseline.csv

Evaluating MobileNetV2 (Std Distill): 100%|██████████| 984/984 [01:27<00:00, 11.26it/s]
```

```
=====
EVALUATION: MobileNetV2 (Std Distill)
=====

Samples:      31474
Overall Acc:   0.9152
Overall AUC:   0.9490
Inference:     0.2 ms/image
Model Size:    9.7 MB (2.6M params)

=====

FAIRNESS REPORT: MobileNetV2 (Std Distill)
=====

Group              Acc      AUC      FPR      TPR      PPV      Count
-----
Female-Asian       0.924   0.961   0.224   0.947   0.964    2931
Female-Black       0.972   0.993   0.098   0.983   0.983     778
Female-Other       0.932   0.967   0.224   0.963   0.956   4563
Female-White       0.913   0.949   0.268   0.943   0.955   9631
Male-Asian         0.891   0.944   0.244   0.926   0.935   1699
Male-Black         0.920   0.878   0.873   0.973   0.944   1015
Male-Other         0.893   0.943   0.199   0.907   0.968   6109
Male-White         0.925   0.943   0.270   0.958   0.955   4748

Fairness Metric              Value      Target      Pass
-----
FFPR Gap (FPR)              0.7748      0.12         X
FEO Gap (Equalized Odds)    0.7748      0.10         X
FDP Gap (Demographic Parity) 0.1837      0.10         X
FOAE Gap (Accuracy)         0.0812      0.08         X
AUC Gap                     0.1149      0.05         X

Worst group (accuracy): Male-Asian
Best group (accuracy):  Female-Black
=====

Results saved to: D:\M3 Projects\DeepFake_Research\outputs\results\MobileNetV2_Std_D
istill.csv

Evaluating MobileNetV2 (Fair Distill): 100%|██████████| 984/984 [01:27<00:00, 11.25i
t/s]
```

```
=====
EVALUATION: MobileNetV2 (Fair Distill)
=====
```

```
Samples:      31474
Overall Acc:   0.9189
Overall AUC:   0.9497
Inference:    0.2 ms/image
Model Size:    9.7 MB (2.6M params)
```

```
=====
FAIRNESS REPORT: MobileNetV2 (Fair Distill)
=====
```

Group	Acc	AUC	FPR	TPR	PPV	Count
Female-Asian	0.921	0.958	0.241	0.947	0.961	2931
Female-Black	0.942	0.978	0.250	0.974	0.959	778
Female-Other	0.941	0.962	0.166	0.962	0.967	4563
Female-White	0.926	0.954	0.205	0.947	0.966	9631
Male-Asian	0.875	0.925	0.303	0.922	0.920	1699
Male-Black	0.945	0.960	0.365	0.965	0.976	1015
Male-Other	0.903	0.942	0.195	0.918	0.969	6109
Male-White	0.910	0.931	0.210	0.930	0.964	4748

Fairness Metric	Value	Target	Pass
FFPR Gap (FPR)	0.1995	0.12	X
FEO Gap (Equalized Odds)	0.1995	0.10	X
FDP Gap (Demographic Parity)	0.1364	0.10	X
FOAE Gap (Accuracy)	0.0702	0.08	✓
AUC Gap	0.0532	0.05	X

Worst group (accuracy): Male-Asian

Best group (accuracy): Male-Black

```
=====
Results saved to: D:\M3 Projects\DeepFake_Research\outputs\results\MobileNetV2_Fair_
Distill.csv
```

✓ In-domain evaluation complete

Cell 9 — Cross-Dataset Evaluation (Celeb-DF & DFD)

```
In [10]: cross_models = {k: v for k, v in models.items() if v is not None}
cross_results = run_full_cross_dataset_evaluation(cross_models, device=DEVICE)
print('\n✓ Cross-dataset evaluation complete')
```

```
=====
CROSS-DATASET EVALUATION: Celeb-DF
=====
```

[Dataset] Loaded 195829 samples for test

REAL: 26665, FAKE: 169164

Evaluating XceptionNet (Teacher) on Celeb-DF: 100%|██████████| 6120/6120 [54:32<00:00, 1.87it/s]

=====

EVALUATION: XceptionNet (Teacher) on Celeb-DF

=====

Samples: 195829
 Overall Acc: 0.8735
 Overall AUC: 0.7398
 Inference: 0.3 ms/image
 Model Size: 83.4 MB (21.9M params)

=====

FAIRNESS REPORT: XceptionNet (Teacher) on Celeb-DF

=====

Group	Acc	AUC	FPR	TPR	PPV	Count
Female-Asian	0.507	0.718	0.789	0.995	0.433	1614
Female-Black	0.136	0.778	0.885	1.000	0.027	169
Female-Other	0.633	0.707	0.890	0.996	0.617	4118
Female-White	0.856	0.745	0.885	0.992	0.859	51446
Male-Asian	0.883	0.720	0.890	0.992	0.887	8196
Male-Black	0.657	0.698	0.837	0.956	0.653	3974
Male-Other	0.858	0.728	0.933	0.995	0.860	19832
Male-White	0.908	0.745	0.901	0.995	0.911	106480

Fairness Metric	Value	Target	Pass
FFPR Gap (FPR)	0.1438	0.12	X
FEO Gap (Equalized Odds)	0.1438	0.10	X
FDP Gap (Demographic Parity)	0.1195	0.10	X
FOAE Gap (Accuracy)	0.7723	0.08	X
AUC Gap	0.0798	0.05	X

Worst group (accuracy): Female-Black

Best group (accuracy): Male-White

=====

Results saved to: D:\M3 Projects\DeepFake_Research\outputs\results\cross_Celeb-DF_XceptionNet (Teacher).csv

=====

CROSS-DATASET EVALUATION: DFD

=====

[Dataset] Loaded 102359 samples for test

REAL: 10735, FAKE: 91624

Evaluating XceptionNet (Teacher) on DFD: 100%|██████████| 3199/3199 [32:08<00:00, 1.66it/s]

```
=====
EVALUATION: XceptionNet (Teacher) on DFD
=====
```

```
Samples:      102359
Overall Acc:   0.8951
Overall AUC:   0.6492
Inference:     0.4 ms/image
Model Size:    83.4 MB (21.9M params)
```

```
=====
FAIRNESS REPORT: XceptionNet (Teacher) on DFD
=====
```

Group	Acc	AUC	FPR	TPR	PPV	Count
Female-Asian	0.900	0.615	0.996	0.998	0.902	2741
Female-Black	0.820	0.686	0.992	1.000	0.820	6677
Female-Other	0.919	0.633	0.995	1.000	0.919	9563
Female-White	0.888	0.651	0.996	0.999	0.889	32866
Male-Asian	0.956	0.744	1.000	0.999	0.957	1807
Male-Black	0.867	0.619	0.999	1.000	0.867	12716
Male-Other	0.927	0.599	0.999	1.000	0.927	16791
Male-White	0.905	0.682	0.995	0.999	0.906	19198

Fairness Metric	Value	Target	Pass
FFPR Gap (FPR)	0.0083	0.12	✓
FEO Gap (Equalized Odds)	0.0083	0.10	✓
FDP Gap (Demographic Parity)	0.0016	0.10	✓
FOAE Gap (Accuracy)	0.1363	0.08	✗
AUC Gap	0.1450	0.05	✗

Worst group (accuracy): Female-Black

Best group (accuracy): Male-Asian

```
=====
Results saved to: D:\M3 Projects\DeepFake_Research\outputs\results\cross_DFD_XceptionNet (Teacher).csv
=====
```

```
=====
CROSS-DATASET EVALUATION: Celeb-DF
=====
```

[Dataset] Loaded 195829 samples for test

REAL: 26665, FAKE: 169164

Evaluating MobileNetV2 (Baseline) on Celeb-DF: 100%|██████████| 6120/6120 [55:22<00:00, 1.84it/s]

```
=====
EVALUATION: MobileNetV2 (Baseline) on Celeb-DF
=====
Samples:      195829
Overall Acc:  0.8732
Overall AUC:  0.7547
Inference:    0.3 ms/image
Model Size:   9.7 MB (2.6M params)
```

```
=====
FAIRNESS REPORT: MobileNetV2 (Baseline) on Celeb-DF
=====
```

Group	Acc	AUC	FPR	TPR	PPV	Count
Female-Asian	0.563	0.756	0.696	0.990	0.463	1614
Female-Black	0.296	0.751	0.721	1.000	0.033	169
Female-Other	0.663	0.756	0.808	0.990	0.638	4118
Female-White	0.858	0.741	0.813	0.982	0.867	51446
Male-Asian	0.882	0.746	0.838	0.984	0.892	8196
Male-Black	0.678	0.736	0.809	0.973	0.665	3974
Male-Other	0.860	0.763	0.877	0.987	0.866	19832
Male-White	0.903	0.754	0.853	0.985	0.915	106480

Fairness Metric	Value	Target	Pass
FFPR Gap (FPR)	0.1811	0.12	X
FEO Gap (Equalized Odds)	0.1811	0.10	X
FDP Gap (Demographic Parity)	0.2441	0.10	X
FOAE Gap (Accuracy)	0.6076	0.08	X
AUC Gap	0.0266	0.05	✓

Worst group (accuracy): Female-Black
Best group (accuracy): Male-White

```
=====
Results saved to: D:\M3 Projects\DeepFake_Research\outputs\results\cross_Celeb-DF_Mo
bileNetV2 (Baseline).csv
```

```
=====
CROSS-DATASET EVALUATION: DFD
=====
[Dataset] Loaded 102359 samples for test
REAL: 10735, FAKE: 91624
```

```
Evaluating MobileNetV2 (Baseline) on DFD: 100%|██████████| 3199/3199 [25:59<00:00,
2.05it/s]
```

```
=====
EVALUATION: MobileNetV2 (Baseline) on DFD
=====
Samples:      102359
Overall Acc:  0.8959
Overall AUC:  0.6920
Inference:    0.2 ms/image
Model Size:   9.7 MB (2.6M params)
```

```
=====
FAIRNESS REPORT: MobileNetV2 (Baseline) on DFD
=====
```

Group	Acc	AUC	FPR	TPR	PPV	Count
Female-Asian	0.902	0.731	0.989	1.000	0.902	2741
Female-Black	0.822	0.763	0.983	1.000	0.821	6677
Female-Other	0.920	0.727	0.982	1.000	0.920	9563
Female-White	0.890	0.686	0.989	1.000	0.890	32866
Male-Asian	0.957	0.712	1.000	1.000	0.957	1807
Male-Black	0.867	0.631	0.999	1.000	0.867	12716
Male-Other	0.928	0.693	0.990	1.000	0.928	16791
Male-White	0.905	0.692	0.997	1.000	0.906	19198

Fairness Metric	Value	Target	Pass
FFPR Gap (FPR)	0.0180	0.12	✓
FEO Gap (Equalized Odds)	0.0180	0.10	✓
FDP Gap (Demographic Parity)	0.0030	0.10	✓
FOAE Gap (Accuracy)	0.1352	0.08	✗
AUC Gap	0.1326	0.05	✗

Worst group (accuracy): Female-Black
Best group (accuracy): Male-Asian

```
=====
```

Results saved to: D:\M3 Projects\DeepFake_Research\outputs\results\cross_DFD_MobileNetV2 (Baseline).csv

```
=====
CROSS-DATASET EVALUATION: Celeb-DF
=====
```

[Dataset] Loaded 195829 samples for test
REAL: 26665, FAKE: 169164

Evaluating MobileNetV2 (Std Distill) on Celeb-DF: 100%6120/6120 [46:29<00:00, 2.19it/s]


```
=====
EVALUATION: MobileNetV2 (Std Distill) on Celeb-DF
=====
Samples:      195829
Overall Acc:  0.8400
Overall AUC:  0.7174
Inference:    0.2 ms/image
Model Size:   9.7 MB (2.6M params)
```

```
=====
FAIRNESS REPORT: MobileNetV2 (Std Distill) on Celeb-DF
=====
```

Group	Acc	AUC	FPR	TPR	PPV	Count
Female-Asian	0.600	0.754	0.595	0.923	0.484	1614
Female-Black	0.172	0.730	0.848	1.000	0.028	169
Female-Other	0.687	0.727	0.722	0.971	0.660	4118
Female-White	0.833	0.750	0.617	0.916	0.889	51446
Male-Asian	0.865	0.709	0.790	0.957	0.895	8196
Male-Black	0.648	0.619	0.659	0.834	0.676	3974
Male-Other	0.831	0.724	0.665	0.917	0.888	19832
Male-White	0.861	0.701	0.684	0.920	0.926	106480

Fairness Metric	Value	Target	Pass
FFPR Gap (FPR)	0.2535	0.12	X
FEO Gap (Equalized Odds)	0.2535	0.10	X
FDP Gap (Demographic Parity)	0.2180	0.10	X
FOAE Gap (Accuracy)	0.6931	0.08	X
AUC Gap	0.1353	0.05	X

Worst group (accuracy): Female-Black
Best group (accuracy): Male-Asian

```
=====
```

Results saved to: D:\M3 Projects\DeepFake_Research\outputs\results\cross_Celeb-DF_MobileNetV2 (Std Distill).csv

```
=====
CROSS-DATASET EVALUATION: DFD
=====
```

[Dataset] Loaded 102359 samples for test
REAL: 10735, FAKE: 91624

Evaluating MobileNetV2 (Std Distill) on DFD: 100% 3199/3199 [23:22<00:00, 2.28it/s]

```
=====
EVALUATION: MobileNetV2 (Std Distill) on DFD
=====
Samples:      102359
Overall Acc:  0.8948
Overall AUC:  0.6011
Inference:    0.2 ms/image
Model Size:   9.7 MB (2.6M params)
```

```
=====
FAIRNESS REPORT: MobileNetV2 (Std Distill) on DFD
=====
```

Group	Acc	AUC	FPR	TPR	PPV	Count
Female-Asian	0.902	0.590	0.996	1.000	0.902	2741
Female-Black	0.819	0.574	0.999	1.000	0.819	6677
Female-Other	0.919	0.666	0.997	0.999	0.919	9563
Female-White	0.888	0.632	0.994	0.999	0.889	32866
Male-Asian	0.957	0.763	1.000	1.000	0.957	1807
Male-Black	0.866	0.523	1.000	1.000	0.867	12716
Male-Other	0.927	0.565	0.998	0.999	0.927	16791
Male-White	0.905	0.635	0.996	0.999	0.906	19198

Fairness Metric	Value	Target	Pass
FFPR Gap (FPR)	0.0060	0.12	✓
FEO Gap (Equalized Odds)	0.0060	0.10	✓
FDP Gap (Demographic Parity)	0.0015	0.10	✓
FOAE Gap (Accuracy)	0.1381	0.08	✗
AUC Gap	0.2400	0.05	✗

Worst group (accuracy): Female-Black
Best group (accuracy): Male-Asian

```
=====
```

Results saved to: D:\M3 Projects\DeepFake_Research\outputs\results\cross_DFD_MobileNetV2 (Std Distill).csv

```
=====
CROSS-DATASET EVALUATION: Celeb-DF
=====
```

[Dataset] Loaded 195829 samples for test
REAL: 26665, FAKE: 169164

Evaluating MobileNetV2 (Fair Distill) on Celeb-DF: 100%|██████████| 6120/6120 [24:14<00:00, 4.21it/s]

```
=====
EVALUATION: MobileNetV2 (Fair Distill) on Celeb-DF
=====
Samples:      195829
Overall Acc:   0.8528
Overall AUC:   0.7470
Inference:     0.2 ms/image
Model Size:    9.7 MB (2.6M params)
```

```
=====
FAIRNESS REPORT: MobileNetV2 (Fair Distill) on Celeb-DF
=====
```

Group	Acc	AUC	FPR	TPR	PPV	Count
Female-Asian	0.584	0.750	0.639	0.951	0.474	1614
Female-Black	0.325	0.795	0.691	1.000	0.034	169
Female-Other	0.679	0.744	0.753	0.979	0.652	4118
Female-White	0.852	0.768	0.647	0.944	0.888	51446
Male-Asian	0.854	0.735	0.689	0.931	0.905	8196
Male-Black	0.674	0.695	0.608	0.846	0.696	3974
Male-Other	0.838	0.718	0.719	0.935	0.882	19832
Male-White	0.874	0.761	0.641	0.930	0.931	106480

Fairness Metric	Value	Target	Pass
FFPR Gap (FPR)	0.1448	0.12	X
FEO Gap (Equalized Odds)	0.1544	0.10	X
FDP Gap (Demographic Parity)	0.2050	0.10	X
FOAE Gap (Accuracy)	0.5489	0.08	X
AUC Gap	0.1002	0.05	X

Worst group (accuracy): Female-Black
Best group (accuracy): Male-White

```
=====
Results saved to: D:\M3 Projects\DeepFake_Research\outputs\results\cross_Celeb-DF_Mo
bileNetV2 (Fair Distill).csv
```

```
=====
CROSS-DATASET EVALUATION: DFD
=====
```

```
[Dataset] Loaded 102359 samples for test
REAL: 10735, FAKE: 91624
```

```
Evaluating MobileNetV2 (Fair Distill) on DFD: 100%|██████████| 3199/3199 [06:03<00:00, 8.79it/s]
```

=====

EVALUATION: MobileNetV2 (Fair Distill) on DFD

=====

Samples: 102359
 Overall Acc: 0.8952
 Overall AUC: 0.7410
 Inference: 0.2 ms/image
 Model Size: 9.7 MB (2.6M params)

=====

FAIRNESS REPORT: MobileNetV2 (Fair Distill) on DFD

=====

Group	Acc	AUC	FPR	TPR	PPV	Count
Female-Asian	0.902	0.723	0.996	1.000	0.902	2741
Female-Black	0.821	0.767	0.984	0.999	0.821	6677
Female-Other	0.919	0.755	0.994	0.999	0.919	9563
Female-White	0.888	0.756	0.989	0.998	0.889	32866
Male-Asian	0.957	0.678	1.000	1.000	0.957	1807
Male-Black	0.866	0.701	0.998	0.999	0.867	12716
Male-Other	0.928	0.674	0.986	1.000	0.928	16791
Male-White	0.905	0.782	0.990	0.999	0.906	19198

Fairness Metric	Value	Target	Pass
FFPR Gap (FPR)	0.0157	0.12	✓
FEO Gap (Equalized Odds)	0.0157	0.10	✓
FDP Gap (Demographic Parity)	0.0034	0.10	✓
FOAE Gap (Accuracy)	0.1360	0.08	✗
AUC Gap	0.1076	0.05	✗

Worst group (accuracy): Female-Black

Best group (accuracy): Male-Asian

Results saved to: D:\M3 Projects\DeepFake_Research\outputs\results\cross_DFD_MobileNetV2 (Fair Distill).csv

=====

CROSS-DATASET SUMMARY

=====

Model	Dataset	AUC	Accuracy	FFPR Gap	FOAE Gap
XceptionNet (Teacher)	Celeb-DF	0.7398	0.8735	0.1438	0.7723
XceptionNet (Teacher)	DFD	0.6492	0.8951	0.0083	0.1363
MobileNetV2 (Baseline)	Celeb-DF	0.7547	0.8732	0.1811	0.6076
MobileNetV2 (Baseline)	DFD	0.6920	0.8959	0.0180	0.1352
MobileNetV2 (Std Distill)	Celeb-DF	0.7174	0.8400	0.2535	0.6931
MobileNetV2 (Std Distill)	DFD	0.6011	0.8948	0.0060	0.1381
MobileNetV2 (Fair Distill)	Celeb-DF	0.7470	0.8528	0.1448	0.5489
MobileNetV2 (Fair Distill)	DFD	0.7410	0.8952	0.0157	0.1360

Saved to: D:\M3 Projects\DeepFake_Research\outputs\results\cross_dataset_summary.csv

✓ Cross-dataset evaluation complete



Cell 10 — Ablation Comparison Table

```
In [14]: rows = []
for name, r in all_results.items():
    fm = r['fairness']
    size_mb = sum(p.nelement()*p.element_size() for p in models[name].parameters())
    rows.append({
        'Method': name,
        'AUC': f"{r['overall_auc']:.4f}",
        'Accuracy': f"{r['overall_accuracy']:.4f}",
        'FFPR Gap ↓': f"{fm['FFPR_gap']:.4f}",
        'FOAE Gap ↓': f"{fm['FOAE_gap']:.4f}",
        'FEO Gap ↓': f"{fm['FEO_gap']:.4f}",
        'FDP Gap ↓': f"{fm['FDP_gap']:.4f}",
        'Size (MB)': f"{size_mb:.1f}",
        'Inference (ms)': f"{r['inference_ms']:.1f}",
    })

df_ablation = pd.DataFrame(rows)

# Style the table
def highlight_best(s):
    is_best = s == s.min()
    return ['background-color: #d4edda; font-weight: bold' if v else '' for v in s]

styled = (df_ablation.style
    .set_caption('📄 Ablation Study — In-Domain (FF++) Results')
    .set_table_styles([{'selector': 'caption', 'props': [('font-size', '14px')], ('font-w
    .apply(highlight_best, subset=['FFPR Gap ↓', 'FOAE Gap ↓', 'FEO Gap ↓', 'FDP Gap ↓
    )
display(styled)

# Save
df_ablation.to_csv(os.path.join(RESULTS_DIR, 'ablation_table.csv'), index=False)
print('Saved: outputs/results/ablation_table.csv')
```

📄 Ablation Study — In-Domain (FF++) Results

	Method	AUC	Accuracy	FFPR Gap ↓	FOAE Gap ↓	FEO Gap ↓	FDP Gap ↓	Size (MB)	Inference (ms)
0	XceptionNet (Teacher)	0.9351	0.9173	0.6373	0.0453	0.6373	0.1164	87.4	0.2
1	MobileNetV2 (Baseline)	0.9521	0.9193	0.5169	0.0663	0.5169	0.1046	10.2	0.2
2	MobileNetV2 (Std Distill)	0.9490	0.9152	0.7748	0.0812	0.7748	0.1837	10.2	0.2
3	MobileNetV2 (Fair Distill)	0.9497	0.9189	0.1995	0.0702	0.1995	0.1364	10.2	0.2

Saved: outputs/results/ablation_table.csv



Cell 11 — Per-Group Accuracy Bar Chart

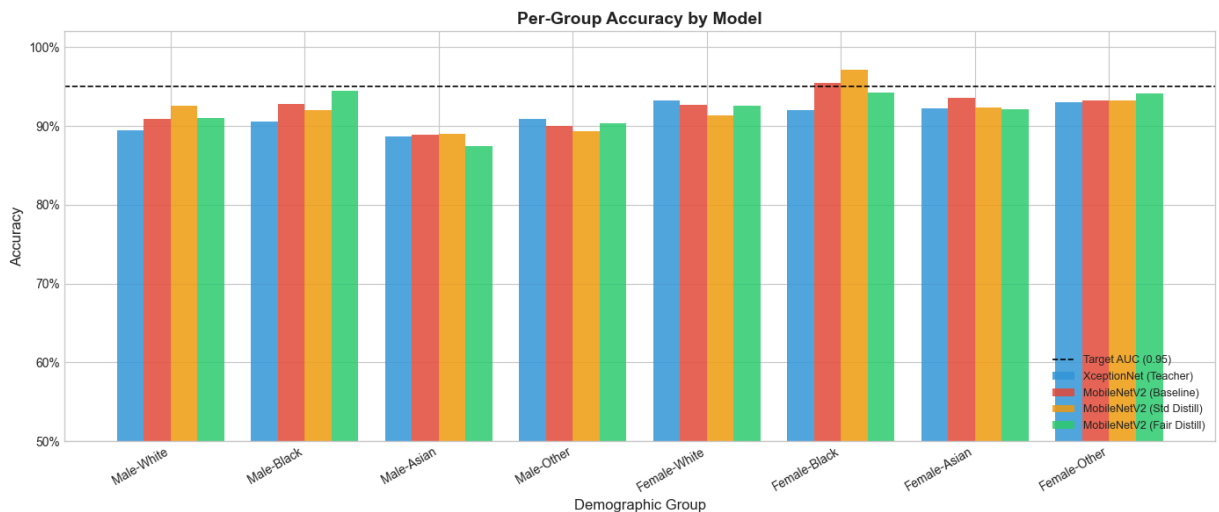
```
In [15]: groups = DEMOGRAPHIC_GROUPS
x = np.arange(len(groups))
width = 0.2
model_names = list(all_results.keys())
bar_colors = ['#3498db', '#e74c3c', '#f39c12', '#2ecc71']

fig, ax = plt.subplots(figsize=(14, 6))
for i, (name, r) in enumerate(all_results.items()):
    accs = [r['fairness']['per_group_accuracy'].get(g, 0) for g in groups]
    bars = ax.bar(x + i*width, accs, width, label=name, color=bar_colors[i % len(bar_colors)])

# Target line
ax.axhline(y=0.95, color='black', linestyle='--', linewidth=1.2, label='Target AUC')

ax.set_xlabel('Demographic Group', fontsize=12)
ax.set_ylabel('Accuracy', fontsize=12)
ax.set_title('Per-Group Accuracy by Model', fontsize=14, fontweight='bold')
ax.set_xticks(x + width * (len(all_results)-1)/2)
ax.set_xticklabels(groups, rotation=30, ha='right')
ax.set_ylim(0.5, 1.02)
ax.legend(loc='lower right', fontsize=9)
ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda y, _: f'{y:.0%}'))

plt.tight_layout()
plt.savefig(os.path.join(FIGURES_DIR, 'per_group_accuracy.png'), dpi=150, bbox_inches='tight')
plt.show()
print('Saved: outputs/figures/per_group_accuracy.png')
```



Saved: outputs/figures/per_group_accuracy.png



Cell 12 — Fairness Gap Summary Chart

```
In [16]: metrics_labels = ['FFPR Gap', 'FOAE Gap', 'FEO Gap', 'FDP Gap']
metric_keys = ['FFPR_gap', 'FOAE_gap', 'FEO_gap', 'FDP_gap']
targets = [0.12, 0.08, 0.10, 0.10]
```

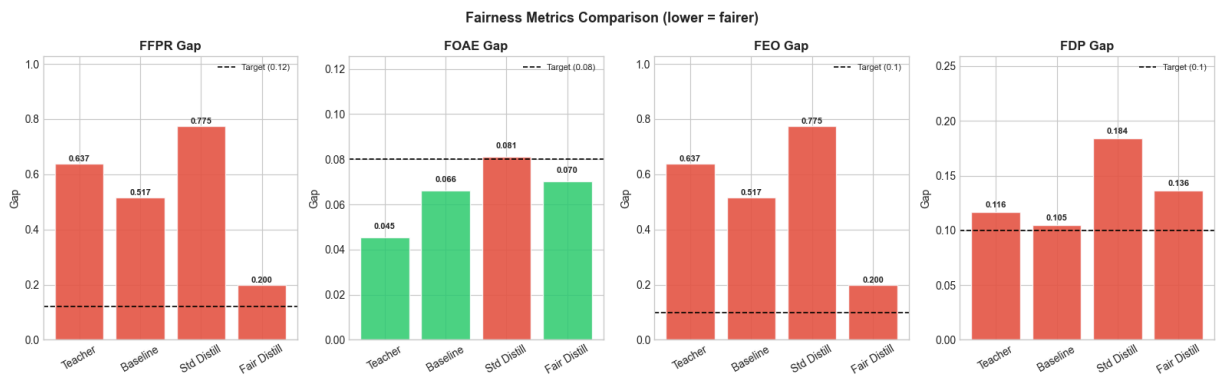
```

fig, axes = plt.subplots(1, len(metrics_labels), figsize=(16, 5))
fig.suptitle('Fairness Metrics Comparison (lower = fairer)', fontsize=13, fontweigh

for ax, label, key, target in zip(axes, metrics_labels, metric_keys, targets):
    vals = [all_results[n]['fairness'][key] for n in all_results]
    names = [n.split('(')[-1].replace(' ', '') for n in all_results]
    colors_bar = ['#2ecc71' if v <= target else '#e74c3c' for v in vals]
    ax.bar(names, vals, color=colors_bar, alpha=0.85, edgecolor='white')
    ax.axhline(y=target, color='black', linestyle='--', linewidth=1.2, label=f'Targ
    ax.set_title(label, fontweight='bold')
    ax.set_ylabel('Gap')
    ax.set_ylim(0, max(vals)*1.3 + 0.02)
    ax.tick_params(axis='x', rotation=30)
    ax.legend(fontsize=8)
    for i, v in enumerate(vals):
        ax.text(i, v + 0.003, f'{v:.3f}', ha='center', va='bottom', fontsize=8, fon

plt.tight_layout()
plt.savefig(os.path.join(FIGURES_DIR, 'fairness_gaps.png'), dpi=150, bbox_inches='ti
plt.show()
print('Saved: outputs/figures/fairness_gaps.png')

```



Saved: outputs/figures/fairness_gaps.png

🔥 Cell 13 — Grad-CAM Heatmaps per Demographic Group

```

In [17]: from src.explainability.gradcam import generate_gradcam_per_group

if fair_model is not None:
    print('Generating Grad-CAM heatmaps for Fair Student...')
    generate_gradcam_per_group(
        fair_model,
        os.path.join(SPLITS_DIR, 'test.csv'),
        num_samples_per_group=3,
        output_dir=GRADCAM_DIR,
        device=DEVICE
    )
    # Display saved heatmaps
    import glob
    from PIL import Image
    heatmap_files = sorted(glob.glob(os.path.join(GRADCAM_DIR, '*.png'))[:8])

```

```

if heatmap_files:
    fig, axes = plt.subplots(2, 4, figsize=(16, 8))
    fig.suptitle('Grad-CAM Heatmaps – Fair Student (per demographic group)', fo
    for ax, fpath in zip(axes.flat, heatmap_files):
        img = Image.open(fpath)
        ax.imshow(img)
        ax.set_title(os.path.basename(fpath).replace('.png', ''), fontsize=8)
        ax.axis('off')
    for ax in axes.flat[len(heatmap_files):]:
        ax.axis('off')
    plt.tight_layout()
    plt.savefig(os.path.join(FIGURES_DIR, 'gradcam_grid.png'), dpi=150, bbox_inc
    plt.show()
else:
    print('[SKIP] Fair student model not loaded.')

```

Generating Grad-CAM heatmaps for Fair Student...

Saved: D:\M3 Projects\DeepFake_Research\outputs\gradcam\gradcam_Female-White.png
 Saved: D:\M3 Projects\DeepFake_Research\outputs\gradcam\gradcam_Female-Other.png
 Saved: D:\M3 Projects\DeepFake_Research\outputs\gradcam\gradcam_Female-Asian.png
 Saved: D:\M3 Projects\DeepFake_Research\outputs\gradcam\gradcam_Male-Asian.png
 Saved: D:\M3 Projects\DeepFake_Research\outputs\gradcam\gradcam_Male-White.png
 Saved: D:\M3 Projects\DeepFake_Research\outputs\gradcam\gradcam_Male-Other.png
 Saved: D:\M3 Projects\DeepFake_Research\outputs\gradcam\gradcam_Male-Black.png
 Saved: D:\M3 Projects\DeepFake_Research\outputs\gradcam\gradcam_Female-Black.png

Grad-CAM visualizations saved to: D:\M3 Projects\DeepFake_Research\outputs\gradcam

⚡ Cell 14 — Model Size & Inference Speed Profiling

```

In [18]: import time

dummy = torch.randn(1, 3, 256, 256).to(DEVICE)
WARMUP = 10
RUNS = 100

rows = []
for name, model in models.items():
    if model is None:
        continue
    model.eval()
    size_mb = sum(p.nelement()*p.element_size() for p in model.parameters()) / 1e6
    params_m = sum(p.numel() for p in model.parameters()) / 1e6

    # Warmup
    with torch.no_grad():
        for _ in range(WARMUP):
            _ = model(dummy)

    # Timed runs
    if DEVICE.type == 'cuda':
        torch.cuda.synchronize()
        t0 = time.perf_counter()
        with torch.no_grad():

```



```

    for _ in range(RUNS):
        _ = model(dummy)
    if DEVICE.type == 'cuda':
        torch.cuda.synchronize()
    elapsed_ms = (time.perf_counter() - t0) / RUNS * 1000

    rows.append({
        'Model': name,
        'Params (M)': f'{params_m:.1f}',
        'Size (MB)': f'{size_mb:.1f}',
        'Inference (ms)': f'{elapsed_ms:.2f}',
        'FPS': f'{1000/elapsed_ms:.0f}',
        '≤15 MB': '✅' if size_mb <= 15 else '❌',
        '≤30 ms': '✅' if elapsed_ms <= 30 else '❌',
    })

df_speed = pd.DataFrame(rows)
display(df_speed.style.set_caption('⚡ Model Efficiency Profiling').hide(axis='index'))
df_speed.to_csv(os.path.join(RESULTS_DIR, 'speed_profiling.csv'), index=False)

# Bar chart
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))
fig.suptitle('Model Efficiency Comparison', fontsize=13, fontweight='bold')

names_short = [n.split('(')[-1].replace(')', '') for n in df_speed['Model']]
sizes = df_speed['Size (MB)'].astype(float)
speeds = df_speed['Inference (ms)'].astype(float)

ax1.barh(names_short, sizes, color=bar_colors[:len(sizes)], alpha=0.85)
ax1.axvline(x=15, color='red', linestyle='--', label='Target (15 MB)')
ax1.set_xlabel('Model Size (MB)')
ax1.set_title('Model Size')
ax1.legend()

ax2.barh(names_short, speeds, color=bar_colors[:len(speeds)], alpha=0.85)
ax2.axvline(x=30, color='red', linestyle='--', label='Target (30 ms)')
ax2.set_xlabel('Inference Time (ms)')
ax2.set_title('Inference Speed')
ax2.legend()

plt.tight_layout()
plt.savefig(os.path.join(FIGURES_DIR, 'efficiency_comparison.png'), dpi=150, bbox_inches='tight')
plt.show()
print('\n✅ All done! Results saved to outputs/results/ and outputs/figures/')

```

⚡ Model Efficiency Profiling

Model	Params (M)	Size (MB)	Inference (ms)	FPS	≤15 MB	≤30 ms
XceptionNet (Teacher)	21.9	87.4	4.47	224	❌	✅
MobileNetV2 (Baseline)	2.6	10.2	5.42	185	✅	✅
MobileNetV2 (Std Distill)	2.6	10.2	7.22	138	✅	✅
MobileNetV2 (Fair Distill)	2.6	10.2	6.74	148	✅	✅

✅ All done! Results saved to outputs/results/ and outputs/figures/

